

API Interface Manual for Control Card Development

(Java)

Version: V1.2.3

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Revised records :

Date	Version	Modified Contents	Author
2019-4-26	V1.0.0	Initial version	
2019-5-8	V1.1.0	Split windows and monitor connection status	
2019-5-24	V1.2.0	Send program by protocol (MultiWindow)	
2019-5-24	V1.2.1	Add send scaled picture	
2019-6-28	V1.2.2	Add sendMulti interface	
2019-8-22	V1.2.3	1. Add timely switch screen interface; 2. Added an interface to exit the multi-window program display. Fix some known issues.	

1、 The interface of send programs

`com.lumen.ledcenter3.protocol.ExtSendUtil``com.lumen.ledcenter3.protocol.ExternalNetworkSendProtocol.OnTcpNetWorkListener`

1.1、 Overview

`ExtSendUtil`

	Name	Description	Complete
1	initNetwork		Completed
2	setListener		Completed
3	splitScreen		Completed
4	sendText		Completed
5	sendPicture		Completed
6	sendPictureWithScale		Completed
7	saveSentDataToFlash		Completed
8	clearSavedDataFromFlash		Completed
9	getBrightness		Completed
10	setBrightness		Completed
11	setScreenOnAndOffTime		Completed
12	getScreenOnAndOffTime		Completed
13	exitSetScreenOnAndOffTime		Completed
14	sendMulti		Completed

OnTcpNetWorkListener

No	Name	Description	Complete
1	onSocketInit		Completed
2	onStatus		Completed
	onBackBytes		Completed
	onTcpProcess		Completed
	breakSocket		Completed

1.2、Detail description

initNetwork

initNetwork (String ip, int port, String idcode)	
Use	
Params	Ip : control card's network ip address Port: net port , default 5200 Idcode: default 255.255.255.255

setListener

setListener (OnTcpNetWorkListener listener)	
Use	
Params	Listener: Monitor the network communicate status and progress

splitScreen

boolean splitScreen(int windowCount, int[]... winRects)	
Use	
Params	windowCount: The number of windows to be split, and within the limits of 1 to 8. winRects : array of windows's rects. winRect is The coordinates of a window. Each window uses four integers to represent its coordinate values of "left, top, right and bottom"
Return	True: successfully , false otherwise.
example	<pre>int[][] winRects = new int[][] {{0,0,32,64},{64,0,96,64}}; extSendUtil.splitScreen(2, winRects);</pre>
notice	This method is time-consuming, you can call it in work thread to avoid blocking of main thread, and monitor the result int OnTcpNetWorkListener.onStatus

sendText

boolean sendText(int nWndNo, String content, int crColor,	
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	int nFontSize, int nSpeed, int nEffect, int nStayTime, int nAlignmentHori, int nAlignmentVert)
Use	
Params	nWndNo: Sequence number of window, virtual value 0-7. Content: text string. crColor: value 1-7 for color: red, green, yellow, blue, purple, blue and white. nFontSize: = 0 means 8 words; = 2 means 16 words; = 3 means 24 words; = 4 means 32 words; = 5 means 40 words; = 6 means 48 words; = 7 means 56 words. nSpeed: value 1 ~ 100. The smaller the value, the faster the speed. nEffect: Display Effect nStayTime: The unit is seconds. nAlignmentHori: Horizontal alignment (0: left alignment, 1: horizontal middle, 2: right alignment) nAlignmentVert: Vertical alignment (0: top alignment, 1: vertical middle, 2: bottom alignment)
Return	True: successfully , false otherwise.
example	<pre>extSendUtil.sendText(0, "欢迎光临123abcef法", 2, 2, 1, ShowEffect.Random.getEffect(), 3, 0, 0);</pre>
notice	This method is time-consuming, you can call it in work thread to avoid blocking of main thread, and monitor the result int OnTcpNetWorkListener.onStatus

sendPicture

```
boolean sendPicture(int nWndNo,
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	int nSpeed, int nEffect, int nStayTime, String picturePath)
Use	
Params	nWndNo: Sequence number of window, virtual value 0-7. nSpeed: value 1 ~ 100. The smaller the value, the faster the speed. nEffect: Display Effect nStayTime: The unit is seconds. picturePath: absolute path of picture file. Include gif, jpg, png.
Return	True: successfully , false otherwise.
example	<pre>extSendUtil.sendPicture(1, 1, ShowEffect.Instant.getEffect(), 3, "C:\\Users\\Administrator\\Desktop\\测试\\turn_circle.gif");</pre>
notice	This method is time-consuming, you can call it in work thread to avoid blocking of main thread, and monitor the result int OnTcpNetWorkListener.onStatus

sendPictureWithScale

	boolean sendPicture(int nWndNo, int winWidth, int winHeight, int nSpeed, int nEffect, int scaleType, int nStayTime, String picturePath)
Use	
Params	nWndNo: Sequence number of window, virtual value 0-7. winWidth: width of window winHeight: height of window nSpeed: value 1 ~ 100. The smaller the value, the faster the speed. nEffect: Display Effect scaleType: picture scale type nStayTime: The unit is seconds. picturePath: absolute path of picture file. Include gif, jpg, png.

Return	True: successfully , false otherwise.
example	<pre>extSendUtil.sendPictureWithScale(1, 96, 64, 1, ShowEffect.<i>Instant</i>.getEffect(), ProtocolConstant.<i>SHOWMODE_TILED</i>, 3, "C:\\Users\\Administrator\\Desktop\\测试\\test_word001_110.jpg");</pre>
notice	This method is time-consuming, you can call it in work thread to avoid blocking of main thread, and monitor the result int OnTcpNetWorkListener.onStatus

sendMulti

boolean sendMulti (int itemNum, String[] texts, int[] colors, int[] fontSizes, int[] showEffects)	
Use	Send multi texts
Params	itemNum : item count texts: text contents colors: text color for each content. value 1-7 for color: red, green, yellow, blue, purple, blue and white. fontSizes: text size for each content showEffects: text showEffect for each content
Return	True: successfully , false otherwise.
example	<pre>extSendUtil.sendMulti(2, new String[]{"happy everyday", "welcome!"}, new int[] {1,2}, new int[] {2,1}, new int[] {ShowEffect.<i>Instant</i>.getEffect(), ShowEffect.<i>Expand_from_bottom</i>.getEffect()});</pre>
notice	This method is time-consuming, you can call it in work thread to avoid blocking of main thread, and monitor the result int OnTcpNetWorkListener.onStatus

saveSentDataToFlash

boolean saveSentDataToFlash()

Use	Save split screen information and programs which sent by sendText and sendPicture. If not call this, the sent data will not be saved when control card restart.
Params	
Return	True: successfully , false otherwise.
notice	This method is time-consuming, you can call it in work thread to avoid blocking of main thread, and monitor the result int OnTcpNetWorkListener.onStatus

clearSavedDataFromFlash

boolean clearSavedDataFromFlash()	
Use	Clear Save split screen information and programs which sent by sendText and sendPicture. And the sent data will not be saved when control card restart.
Params	
Return	True: successfully , false otherwise.
notice	This method is time-consuming, you can call it in work thread to avoid blocking of main thread, and monitor the result int OnTcpNetWorkListener.onStatus

getBrightness

int[] getBrightness()	
Use	
Params	
Return	The brightness value, length is 24, represent 24 time slot from 00:00.

notice	This method is time-consuming, you can call it in work thread to avoid blocking of main thread, and monitor the result int OnTcpNetWorkListener.onStatus
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setBrightness

boolean setBrightness(int[] brigghtnessValue)	
Use	
Params	brigghtnessValue: The brightness value, length is 24, represent 24 time slot from 00:00. When all 24 value is -1, then is in photosensitivity mode.
Return	True: successfully , false otherwise.
notice	This method is time-consuming, you can call it in work thread to avoid blocking of main thread, and monitor the result int OnTcpNetWorkListener.onStatus

setScreenOnAndOffTime

boolean setScreenOnAndOffTime(int nHourOn, int nMinuteOn, int nHourOff, int nMinuteOff)	
Use	
Params	nHourOn: 0~23 nMinuteOn: 0~59 nHourOff: 0~23 nMinuteOff: 0~59
Return	True: successfully , false otherwise.
notice	This method is time-consuming, you can call it in work thread to avoid blocking of main thread, and monitor the result int OnTcpNetWorkListener.onStatus

getScreenOnAndOffTime

int[] getScreenOnAndOffTime ()	
Use	
Params	
return	The screen on and off time, length is 4. When value is 255, explain that the switch screen settings are not turned on.
notice	This method is time-consuming, you can call it in work thread to avoid blocking of main thread, and monitor the result int OnTcpNetWorkListener.onStatus

exitSetScreenOnAndOffTime

boolean exitSetScreenOnAndOffTime ()	
Use	Exit screen on and off settings.
Params	
Return	True: successfully , false otherwise.
notice	This method is time-consuming, you can call it in work thread to avoid blocking of main thread, and monitor the result int OnTcpNetWorkListener.onStatus

setScreenSwitch

boolean setScreenSwitch(boolean turnOn)	
Use	Switch the screen immediately (restart card is invalid)
Params	TurnOn : true means ON , false means OFF
Return	True: successfully , false otherwise.

notice	This method is time-consuming, you can call it in work thread to avoid blocking of main thread, and monitor the result int_ OnTcpNetWorkListener.onStatus
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quitExternalScreen

boolean quitExternalScreen()	
Use	Exit multi-window program mode and return to regular program playback
Params	
Return	True: successfully , false otherwise.
notice	This method is time-consuming, you can call it in work thread to avoid blocking of main thread, and monitor the result int_ OnTcpNetWorkListener.onStatus

OnTcpNetWorkListener

onSocketInit

onSocketInit(int result);	
Use	Monitor socket connect status
Params	Result: 0 fail,1 success

onStatus

onStatus(int status, int socketindex);	
Use	Monitor data send status
Params	Status: -1 means error occur , 1 means ok. Socketindex: socket Index

onBackBytes

onBackBytes(int[] backbytes, int socketindex);	
Use	
Params	Backbytes: back data from control card Socketindex : socket Index

onTcpProcess

onTcpProcess(int process, int totalprocess, int socketindex);	
Use	Monitor data send progress when send picture or other files
Params	Process : Totalprocess : Socketindex : socket Index

breakSocket

breakSocket(int socketindex);	
Use	Monitor socket closed.
Params	Socketindex: socket Index

1.3、Demo

2、 Supplementary to parameter description

2.1、 Color and grey scale

Name	Value	Constant name (in ProtocolConstant class)
Mono	0x01	COLOR_TYPE_MONO
Mono with 256-grade grey scale	0x71	COLOR_TYPE_MONO_WITH_GRAY
Dual colors	0x03	COLOR_TYPE_RG_COLOR
Dual colors with 256-grade grey scale	0x73	COLOR_TYPE_RG_COLOR_WITH_GRAY
RGB color	0x07	COLOR_TYPE_RGB_COLOR
Full color	0x77	COLOR_TYPE_FULL_COLOR

2.2、 Text and Picture Display Effect

example : ShowEffect.Random.getEffect();

Random
Instant
Open_left
Open_right
Open_horizontal
Open_vertical
Shutter_vertical
Shift_left
Shift_right

Shift_up
Shift_down
Scroll_up
Scroll_left
Scrollleft_continuously
Scroll_right
Scroll_right_continuously
Blink
Shutter_horizontal
Open_clockwise
Open_anticlockwise
Windmill_clockwise
Windmill_anticlockwise
Rectangle_out
Rectangle_in
Corner_out
Corner_in
Round_out
Round_in
Open_top_left
Open_top_right

Open_bottom_left
Open_bottom_right
Open_slash
Open_backslash
Slide_top_left
Slide_top_right
Slide_bottom_left
Slide_bottom_right
Open_cross_in
Open_cross_out
Zebra_cross_horizontal
Zebra_cross_vertical
Mosaic_large
Mosaic_small
Laser_line_upward
Laser_line_downward
Scrape_up
Drop_down
Slide_left_right
Slide_top_bottom
Slewing_out

Slewing_in
Chessboard_horizontal
Chessboard_vertical
Scroll_up_continuously
Scroll_down_continuously
Expand_from_top
Expand_from_bottom
Expand_vertical
Blind_horizontal
Blind_vertical
Snow_fall
Scroll_down
Open_left_right
Open_up_down
Open_2_fan
Slide_zebra_horizontal
Slide_zebra_vertical

2.3、picture scale type

Value	Type	Constant name (ProtocolConstant)
0	Center	SHOWMODE_CENTER
1	Scale	SHOWMODE_SCALE

2	Stretch	SHOWMODE _STRETCH
3	Tiled	SHOWMODE _TILED

2.4、font size

Note: The dot matrix size of extended fonts supported by each control card is different. If the text is not displayed in a certain size, the dot matrix size is not supported.

Value	Size (pixel)	Constant name (ProtocolConstant)
0	8	FONTSIZE_8
1	12	FONTSIZE_12
2	16	FONTSIZE_16
3	24	FONTSIZE_24
4	32	FONTSIZE_32
5	40	FONTSIZE_40
6	48	FONTSIZE_48
7	56	FONTSIZE_56

2.5、socket Index

```
SOCKETINDEX_SPLIT_SCREEN = 2;  
SOCKETINDEX_SEND_TEXT = 3;  
SOCKETINDEX_SEND_PICTURE = 4;  
SOCKETINDEX_SAVE_DATA = 5;  
SOCKETINDEX_CLEAR_DATA = 6;  
SOCKETINDEX_GET_BRIGHTNESS = 7;  
SOCKETINDEX_SET_BRIGHTNESS = 8;  
SOCKETINDEX_SET_SCREEN_ONOFF = 9;  
SOCKETINDEX_GET_SCREEN_ONOFF = 10;  
SOCKETINDEX_EXIT_SET_SCREEN_ONOFF = 11;
```